Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2009-I-Q03] | Part A (Q1-Q5)

A coin has head probability $0 < p < 1$. In eight independent flips, the probability of three heads is one twenty-fifth of that of five heads. Write $p = m/n$ reduced and find $m + n$.

硬币正面概率为 $0 < p < 1$。独立掷八次，恰三正面的概率为恰五正面概率的二十五分之一。若 $p = m/n$ 最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2010-I-Q04] | Part A (Q1-Q5)

Two players independently toss the same three coins once each. Two coins are fair; the third has head probability $4/7$. Their head counts agree with reduced probability m/n. Find $m+n$.

两名玩家各独立投掷三枚硬币一次；其中两枚公平，另一枚正面概率为 $4/7$。两人正面数相同的概率为最简分数 m/n，求 $m+n$。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2006-II-Q10] | Part B (Q6-Q10)

Seven teams play an independent fair round robin without draws. Given that team A beats B in their game, the chance that A ends with more wins is reduced m/n. Find $m+n$.

七队进行无平局、各场独立公平的单循环赛。已知 A 战胜 B，最终 A 胜场数更多的概率为最简分数 m/n，求 $m+n$。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2008-I-Q09] | Part B (Q6-Q10)

Ten independent rectangular crates have dimensions $3, 4, 6$ feet, with each possible vertical dimension equally likely. The probability their total stacked height is 41 feet is reduced m/n. Find m.

十个长方体箱尺寸均为三、四、六英尺，各自独立等可能选择一种尺寸为高度，叠高恰四十一英尺的概率为最简分数 m/n，求 m。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2009-II-Q08] | Part B (Q6-Q10)

Two players independently roll fair dice until their first six. If the probability their roll counts differ by at most one is reduced m/n, find $m+n$.

两人独立掷公平骰直至第一次出现六。掷次相差至多一次的概率为最简分数 m/n，求 $m+n$。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2021-II-Q08] | Part B (Q6-Q10)

An ant begins at a bottom vertex of a cube. Its first step chooses uniformly among the three edges. Later it chooses uniformly among the two edges that do not return immediately to the previous vertex. The chance of being on the top face after eight steps is m/n reduced. Find $m+n$.

蚂蚁从立方体底面顶点出发，第一步等概率选三条边之一。以后每步从不立即返回前一顶点的两条边中等概率选择。八步后在顶面的概率约分为 m/n，求 $m+n$。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2001-II-Q11] | Part C (Q11-Q15)

A team plays six independent matches, each equally likely to be a win, loss, or tie. If the probability of more wins than losses is m/n reduced, find $m+n$.

某队独立比赛六场，每场胜、负、平概率均为 $1/3$。若胜场多于负场的概率约分为 m/n，求 $m+n$。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2002-II-Q12] | Part C (Q11-Q15)

Each independent basketball shot succeeds with probability $2/5$. After ten shots, require exactly four successes and no prefix of one through nine shots having success fraction above $2/5$. The probability is $p^a q^b r / s^c$, where p, q, r, s are primes and $a, b, c > 0$. Find $(p + q + r + s)(a + b + c)$.

每次投篮独立命中，概率为 $2/5$。要求十次后恰中四次，且前一至九次任一前缀命中率都不超过 $2/5$。将概率写为 $p^a q^b r / s^c$，各底数为质数、指数为正，求 $(p + q + r + s)(a + b + c)$。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2003-II-Q13] | Part C (Q11-Q15)

A bug starts at a triangle vertex. On each of ten steps, it independently chooses either other vertex with equal probability. If its return probability after the tenth step is m/n reduced, find $m+n$.

虫从三角形一顶点出发，每步等概率选另外两顶点之一前往。十步后回到起点的概率约分为 m/n，求 $m+n$。

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Section 5.5 Distributions & repeated trials

分布与重复试验

[AIME-2014-I-Q11] | Part C (Q11-Q15)

A token makes six independent uniform steps on the integer grid from the origin, each one unit north, south, east, or west. Its chance of finishing on $|x| = |y|$ is reduced m/n. Find $m + n$.

棋子从原点独立等可能地向上下左右各一步，连续六步。终点落在 $|x| = |y|$ 上的概率为最简分数 m/n，求 $m + n$。

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